

Finite-state outcome and strategy algorithms for a no-echo subtraction game

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Abstract

We consider the one-heap subtraction game in which a move removes between one and m stones, but may not repeat the opponent's immediately preceding take. A player with no legal move loses, even when stones remain. We give a row recurrence that determines the outcome and a winning move at every state, including states with a specified preceding take. It uses at most nm candidate checks for a heap of size n . For each fixed m , the recurrence also yields a finite-state proof of eventual periodicity. For the parameters $m + 1 = 2h$ with odd 2-adic valuation, we use Gurvich's two-gap theorem to construct a smaller deterministic profile on h bits. After the first profile, at most 2^{h-2} profiles are reachable. Its eventual cycle gives an exact prefix-and-offset representation of all losing opening heaps, permitting binary-index outcome queries and a winning opening move after finite preprocessing. The construction does not give a closed formula for the cycle as a function of m .

Keywords: impartial games; subtraction games; finite-state recurrence; periodicity; winning strategy.

1 Introduction

The game D.U.D.E.N.E.Y. restricts a one-heap subtraction game by disallowing the number of stones taken on the immediately previous turn. It was discussed in the combinatorial-games literature and later studied by Gurvich [1, 2]. The previous take is part of the state: an opening heap of size n and an in-play heap of size n need not have the same outcome.

Gurvich writes $k = m + 1$. His theorem classifies the losing opening heaps when $v_2(k)$ is even: they are the multiples of k . When $v_2(k)$ is odd, he proves that the first positive losing opening heap is $k + 1$ and that consecutive losing heaps differ by k or $k + 1$; he leaves the selection of those gaps open [1]. The later problem list still describes the odd-cap analysis as partial [2].

There are two distinct questions. The first asks for an algorithm that computes the outcome and a winning move for any specified finite state. The second asks for a uniform arithmetic description of all losing opening heaps in terms of the parameters. We address the first question and give an explicit finite-state description for each fixed cap. The state recurrence in Section 2 is a direct dynamic-programming form of minimax. The additional profile construction in Section 3 uses the two-gap structure to replace a generic row window with a smaller state space.

2 Rules and an exact row recurrence

Fix an integer $m \geq 1$. A state is (n, j) , where $n \geq 0$ is the number of stones remaining and $j \in \{1, \dots, m\}$ records the last take. For an opening state we write $j = \perp$. The legal takes from

(n, j) are

$$L(n, j) = \{i : 1 \leq i \leq \min(m, n), i \neq j\}, \quad L(n, \perp) = \{i : 1 \leq i \leq \min(m, n)\}.$$

Taking i leads to $(n - i, i)$. We use normal play with the convention that a player without a legal take loses, including when $n > 0$. Since every move decreases n , every state has a well-defined outcome: it is **P** if the player to move loses under optimal play and **N** otherwise.

Definition 2.1. For $n \geq 0$, let T_n be the set of takes that would move an opening heap of size n to a **P** state:

$$T_n = \{i : 1 \leq i \leq \min(m, n), (n - i, i) \text{ is } \mathbf{P}\}.$$

In particular $T_0 = \emptyset$.

Lemma 2.2 (State criterion). *For every $n \geq 0$ and $j \in \{1, \dots, m\}$,*

$$(n, j) \text{ is } \mathbf{P} \iff T_n \subseteq \{j\}. \quad (2.1)$$

*The opening state $(n, \perp)$ is **P** if and only if $T_n = \emptyset$. When a state is **N**, any $i \in T_n$ with $i \neq j$ is a winning take; for $j = \perp$, any $i \in T_n$ is a winning take.*

Proof. The legal moves from (n, j) to **P** states are exactly $T_n \setminus \{j\}$. A finite impartial state is **P** precisely when none of its legal moves has a **P** child. Thus it is **P** precisely when $T_n \setminus \{j\} = \emptyset$, which is equivalent to (2.1). With $j = \perp$ nothing is excluded. The remaining assertions follow from the same description of the winning takes. $\square$

Theorem 2.3 (Exact all-state recurrence). *For $n \geq 1$,*

$$T_n = \{i : 1 \leq i \leq \min(m, n), T_{n-i} = \emptyset \text{ or } T_{n-i} = \{i\}\}. \quad (2.2)$$

*Starting with $T_0 = \emptyset$, equation (2.2) computes every T_n uniquely. After computing $T_0, \dots, T_N$, one can decide every state (n, j) with $n \leq N$ and choose a winning take at every **N** state. The calculation uses at most Nm candidate checks and $O(Nm)$ membership bits if all sets are retained.*

Proof. Taking i from a heap of size n reaches $(n - i, i)$. By Lemma 2.2, this child is **P** exactly when $T_{n-i} \subseteq \{i\}$. A set satisfying this inclusion is either empty or the singleton $\{i\}$, proving (2.2). Each right-hand row has smaller index, so induction from T_0 gives a unique answer for every n . Lemma 2.2 then supplies all outcomes and moves. There are at most m candidate takes per row and each row is a subset of $\{1, \dots, m\}$, giving the stated bounds. The bound counts candidate checks; it is not a bit-complexity claim for binary-encoded N . $\square$

The terminal case with stones remaining is automatically included. For example, $T_1 = \{1\}$, hence $(1, 1)$ is **P** because its only possible take is forbidden, while $(1, \perp)$ is **N**. If $m = 1$, all positive opening heaps are **N**: the opening take of one stone is followed by a state with no legal move whenever any stones remain.

2.1 A generic finite-state consequence

Define a compressed symbol

$$c_n = \begin{cases} 0, & T_n = \emptyset, \\ i, & T_n = \{i\}, \\ m + 1, & |T_n| \geq 2. \end{cases}$$

For each candidate i , recurrence (2.2) asks only whether $c_{n-i} = 0$ or $c_{n-i} = i$. Consequently c_n is determined by the preceding m symbols once $n \geq m$. There are at most $(m+2)^m$ such windows. Two of the first $(m+2)^m + 1$ windows coincide; determinism then makes the subsequent symbol sequence periodic. The opening P/N sequence and the outcome sequence for each fixed last take are therefore eventually periodic for every fixed m . This elementary finite-state argument does not identify a shortest period or a symbolic period formula in m .

3 Dopey parameters and the gap selector

Put $K = m + 1 = 2h$ and assume $v_2(K)$ is odd. We take $m \geq 2$, so $h \geq 3$. Let

$$0 = p_0 < p_1 < p_2 < \cdots$$

be the opening P heaps. We use the following result of Gurvich [1]:

$$p_1 = K + 1, \quad g_r := p_{r+1} - p_r \in \{K, K + 1\} \quad (r \geq 0). \quad (3.1)$$

This is the only external game-theoretic theorem used in the profile construction.

Lemma 3.1 (Forced take). *If $T_p = \emptyset$ and $1 \leq d \leq m$, then $d \in T_{p+d}$. If $p + d$ is strictly before the next opening P heap, a singleton T_{p+d} must therefore be $\{d\}$.*

Proof. Taking d from $p + d$ leaves (p, d) . This is P by Lemma 2.2, since T_p is empty. The last assertion follows immediately. $\square$

Lemma 3.2 (Midpoint selector). *For every $r \geq 0$,*

$$g_r = K + 1 \iff T_{p_r+h} = \{h\}. \quad (3.2)$$

If $g_r = K + 1$, then $T_{p_r+K} = \{h\}$.

Proof. Let $p = p_r$. In recurrence (2.2) for T_{p+K} , a candidate i has predecessor heap $p + d$ where $d = K - i \in \{1, \dots, m\}$. By (3.1) this predecessor is before the next opening P heap, hence its row is nonempty and contains d by Lemma 3.1. It can equal the singleton $\{i\}$ only if $i = d = h$. Thus T_{p+K} is $\{h\}$ when $T_{p+h} = \{h\}$ and is empty otherwise. The two-gap theorem (3.1) converts this empty/nonempty test into (3.2). $\square$

For $1 \leq d \leq m$ define the boundary singleton bit

$$s_r(d) = \begin{cases} 1, & T_{p_r+d} = \{d\}, \\ 0, & \text{otherwise.} \end{cases} \quad (3.3)$$

By Lemma 3.2, $g_r = K + s_r(h)$.

Lemma 3.3 (Boundary row formula). *For the initial boundary $p_0 = 0$, the only possible members of T_d are d and, when d is even, $d/2$; the latter occurs precisely when $s_0(d/2) = 1$.*

For $r \geq 1$, write $p = p_r$, $p^- = p_{r-1}$ and $g = p - p^- \in \{K, K + 1\}$. For $1 \leq d \leq m$, the row T_{p+d} contains d , and its only other possible members are:

1. $d/2$, if d is even and $s_r(d/2) = 1$;
2. $e = (g + d)/2$, if $g - d$ is even, $e \leq m$, and $s_{r-1}(e) = 1$;

3. h , if $g = K + 1$ and $d = h - 1$.

Each listed condition is also sufficient for the corresponding take to belong to the row.

Proof. The take d is forced by Lemma 3.1. Consider another candidate take $i \neq d$, whose predecessor heap is $q = p + d - i$. If $i < d$, then $q = p + (d - i)$ lies after p and before the next opening P heap. The row T_q contains $d - i$ and is nonempty. For i to enter T_{p+d} , recurrence (2.2) requires $T_q = \{i\}$, so $i = d - i = d/2$. This occurs exactly when d is even and $s_r(d/2) = 1$. When $r = 0$, no candidate $i > d$ is legal, proving the initial assertion.

Now suppose $r \geq 1$ and $i > d$. Set $t = i - d$, so $q = p - t = p^- + (g - t)$. If $g - t \leq m$, the row T_q contains the forced take $g - t$ to p^- . It equals $\{i\}$ only when $g - t = i = d + t$, which forces $g - d = 2t$ and $i = (g + d)/2 = e$. The parity and range conditions in item 2 are exactly those making this candidate legal. When they hold, the singleton condition is $s_{r-1}(e) = 1$.

The remaining possibility $g - t > m$ requires $g = K + 1 = m + 2$ and $t = 1$; then $q = p - 1 = p^- + K$. By Lemma 3.2, $T_q = \{h\}$. Therefore $i = d + 1$ enters the row just when $d + 1 = h$, which is item 3. These cases exhaust the candidates. $\square$

3.1 The profile transition

Let $H_r = (s_r(h), s_r(h+1), \dots, s_r(m)) \in \{0, 1\}^h$. Given H_r , write $\delta = H_r[0] = s_r(h)$ and $g = K + \delta$. Define $x_1, \dots, x_m$ in increasing order by

$$x_d = 1 \iff \begin{cases} d \text{ is odd, or } x_{d/2} = 0 \text{ if } d \text{ is even;} \\ \text{there is no } e = (g + d)/2 \leq m \text{ with } g - d \text{ even and } H_r[e - h] = 1; \\ \neg(\delta = 1 \text{ and } d = h - 1). \end{cases} \quad (3.4)$$

In the second condition, the indexed e lies in $\{h, \dots, m\}$; indeed it is greater than h whenever the parity condition holds. The boundary formula proves that $x_d = s_{r+1}(d)$ and therefore

$$H_{r+1} = F(H_r) := (x_h, \dots, x_m). \quad (3.5)$$

The same-boundary reference $x_{d/2}$ has smaller index, so this is a deterministic computation taking $O(m)$ Boolean steps.

For $r = 0$, Lemma 3.3 shows that

$$s_0(d) = 1 \iff v_2(d) \text{ is even} \quad (1 \leq d \leq m). \quad (3.6)$$

Indeed, odd d have no additional candidate, whereas an even d has the additional take $d/2$ exactly when $s_0(d/2) = 1$. This alternates under halving. Equation (3.6) initializes H_0 without computing the heap rows.

Lemma 3.4 (Tail-state bound). *Every reachable upper profile has $s_r(m) = 1$. For $r \geq 0$ its successor satisfies*

$$(s_{r+1}(m-2), s_{r+1}(m-1), s_{r+1}(m)) = (1 - \delta, \delta, 1), \quad \delta = s_r(h). \quad (3.7)$$

Thus the profiles H_r with $r \geq 1$ occupy at most 2^{h-2} states.

Proof. Since $m = 2h - 1$ is odd, (3.6) gives $s_0(m) = 1$. For a successor, apply (3.4) at $d = m$: this d is odd, there is no legal old-profile extra, and the exceptional condition is absent. Hence $s_{r+1}(m) = 1$.

At $d = m - 1 = 2h - 2$, if $\delta = 0$, the old extra is $e = m$ and is present because $s_r(m) = 1$; thus the successor bit is zero. If $\delta = 1$, there is no old extra. The same-boundary candidate is governed by x_{h-1} , but $x_{h-1} = 0$ because the exceptional condition is active there. Thus the bit is one. At $d = m - 2 = 2h - 3$, the same-boundary candidate is absent by oddness. When $\delta = 0$ the old extra is absent, giving one; when $\delta = 1$ the old extra is $e = m$, giving zero. This proves (3.7). The three coordinates in (3.7) belong to H_{r+1} because $h \geq 3$. Their two possible patterns leave $h - 3$ unrestricted coordinates, hence at most $2 \cdot 2^{h-3} = 2^{h-2}$ states. $\square$

Theorem 3.5 (Eventually periodic gap representation). *Let $N_h = 2^{h-2}$. Starting from $H_1 = F(H_0)$, iterate F until the first repeated profile. There are $\mu \geq 0$ and $\lambda \geq 1$ with $\mu + \lambda \leq N_h$ and*

$$H_{1+\mu+t+\lambda} = H_{1+\mu+t} \quad (t \geq 0).$$

Put $r_* = 1 + \mu$, $P_* = p_{r_*}$,

$$C_0 = 0, \quad C_t = \sum_{j=0}^{t-1} (K + H_{r_*+j}[0]) \quad (1 \leq t \leq \lambda), \quad L = C_\lambda.$$

Then the full set of losing opening heaps is

$$\{p_0, \dots, p_{r_*-1}\} \cup \{P_* + qL + C_t : q \geq 0, 0 \leq t < \lambda\}. \quad (3.8)$$

All quantities in (3.8) are constructed by a finite algorithm, with $O(KN_h)$ Boolean profile work and finite additions.

Proof. Lemma 3.4 confines $H_1, H_2, \dots$ to a set of at most N_h states. Among the first $N_h + 1$ profiles, one repeats. Because F is deterministic, a repeated profile has the same entire future. The output gap at H_r is $K + H_r[0]$ by Lemma 3.2. Thus gaps repeat from index r_* with period λ . Their cumulative offsets within one block are the C_t , and the total block displacement is L . The earlier heaps $p_0, \dots, p_{r_*-1}$ form the finite prefix, proving (3.8). Generating each profile uses $O(K)$ Boolean work, and at most N_h tail profiles are generated before the repetition is certified. $\square$

Corollary 3.6 (Opening query and strategy). *After the preprocessing in Theorem 3.5, an arbitrary opening heap n can be classified by a finite-prefix search when $n < P_*$ and, otherwise, by division with remainder by L and a search among the offsets $C_0, \dots, C_{\lambda-1}$. A winning opening take can be returned whenever the opening state is $\mathbf{N}$.*

Proof. The prefix and the periodic offsets in (3.8) are strictly ordered because all gaps are positive, so the stated membership query is exact. If n is not a $\mathbf{P}$ heap, let $p_r < n < p_{r+1}$ be its preceding $\mathbf{P}$ heap and put $d = n - p_r$. When $d \leq m$, take d : the child (p_r, d) is $\mathbf{P}$ by Lemma 2.2. The only other possibility is $d = K$, in a long gap $g_r = K + 1$. Then take h ; Lemma 3.2 gives $T_{p_r+h} = \{h\}$, so the child $(p_r + h, h)$ is $\mathbf{P}$ by Lemma 2.2. $\square$

For a B -bit input n , the query following preprocessing uses binary division by the $O(h + \log K)$ -bit value L and binary search among at most N_h offsets. In particular it depends polynomially on B for each fixed m . The preprocessing is exponential in h , so this is not a polynomial-time bound in the joint binary input lengths of n and m .

4 Example and scope

For $m = 5$, $K = 6$ and $h = 3$. Formula (3.6) gives $H_0 = (1, 1, 1)$. The transition yields $H_1 = (0, 1, 1)$ and $H_2 = (1, 0, 1)$, which alternate. The first gap is 7, followed by a repeating pair of gaps 6, 7. Thus $r_* = 1$, $P_* = 7$, $L = 13$, and (3.8) becomes

$$\{0\} \cup \{7 + 13q : q \geq 0\} \cup \{13 + 13q : q \geq 0\}.$$

This illustrates the need to allow an initial transient: the first gap is not part of the tail's cyclic ordering.

The row recurrence gives an outcome and a legal winning move from every specified finite state (n, j) ; in this precise all-state sense it is an exact strategy solution. The profile theorem gives an explicit finite representation for each fixed dopey cap and a faster query in the heap size after preprocessing. Neither result supplies a simple arithmetic formula for the gap word, its least period, or its preperiod as a function of the varying cap m . In particular, eventual periodicity here should not be conflated with pure periodicity from the first gap. The broader uniform classification problem recorded in [1, 2] remains outside the claims of this article.

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